






Time Series Analysis

Chapter 12: Spectral analysis



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Acronyms used in this material

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **CWT**: Continuous Wavelet Transform
- **DFT**: Discrete Fourier Transform
- **ENTSO-E**: European Network of Transmission System Operators for Electricity
- **EODHD**: EOD Historical Data (market-data provider)
- **EU**: European Union
- **FFT**: Fast Fourier Transform
- **FRED**: Federal Reserve Economic Data
- **GDP**: Gross Domestic Product
- **GPH**: Geweke–Porter-Hudak (estimator)
- **HP**: Hodrick–Prescott (filter)
- **LPPL**: Log-Periodic Power Law
- **MA**: Moving Average
- **OLS**: Ordinary Least Squares
- **SILSO**: Sunspot Index and Long-term Solar Observations
- **STFT**: Short-Time Fourier Transform
- **US**: United States
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)

The chapter's question and route

- **Question**: which cycles does a series contain, and how much of its variance does each cycle explain?
  - examples: the 11-year sunspot cycle, the business cycle, the daily and weekly rhythm of electricity demand
- **Route** of the chapter
  - cycles, frequencies and the discrete Fourier transform
  - the spectral density of white noise and of ARMA models
  - the periodogram, its weaknesses, and how to smooth it
  - cycles in data; long memory; filters; two series; wavelets as a pointer
- Self-study chapter: worked examples, interpretation slides and a recap after each section; self-assessment with answers at the end

Learning outcomes

- Convert between frequency, period and angular frequency, and find the Fourier frequencies of a sample
- Compute the spectral density of white noise, AR(1), MA(1) and AR(2) models and read its shape
- Compute a periodogram by hand for a short series and explain why the raw periodogram is not consistent
- Smooth the periodogram (Daniell, Bartlett, Welch), taper the data and build chi-square confidence bands
- Identify cycles in sunspots, GDP and electricity load; read coherence and phase between two series

Reading and tools

- Main reference for this chapter: Shumway and Stoffer (2017), Ch. 4 (spectral analysis and filtering)
  - theory: Brockwell and Davis (1991), Ch. 4 and 10; Hamilton (1994), Ch. 6; applications in Python: Huang and Petukhina (2022)
  - Shumway and Stoffer (2017) measure frequency in cycles per observation; Chapter 8 used the angular frequency $\lambda = 2\pi\nu$
- Python Quantlets of this chapter: Quantlets/Ch_12
  - the periodogram, the Daniell smoother and the Morlet wavelet are written out in `numpy`; Welch and coherence come from `scipy.signal`
- Lecture notebook: open in Google Colab
- Video course: Applied Time Series Analysis with Python

Joseph Fourier and the idea of the chapter



J. B. J. Fourier (1768–1830)

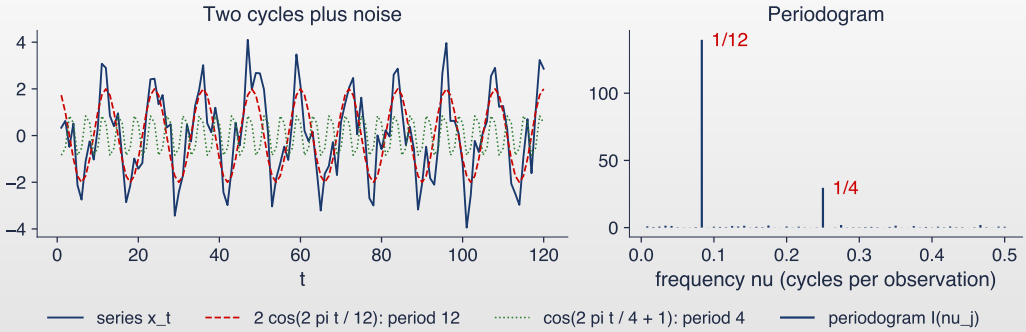
Engraving: A. F. B. Geille, after J.-L. Boilly (1839–1840); public domain; Wikimedia Commons

- Fourier (1807, heat conduction): a function can be written as a sum of sines and cosines
  - each term is a pure cycle with its own frequency and amplitude
- Applied to a time series: how much variance sits at each frequency?
  - the **time domain** (Chapters 1–8) describes the series through its autocorrelations
  - the **frequency domain** describes the same information through the spectrum
- The two views are equivalent; the spectrum makes cycles visible at a glance

A cycle: amplitude, frequency, phase

- A pure cycle: $x_t = A\cos(2\pi\nu t + \varphi)$, $t = 1, 2, \ldots$
  - A: **amplitude** (height of the wave); φ: **phase** (where the wave starts)
  - ν: **frequency**, in cycles per observation; the **period** $1/\nu$ is the number of observations per cycle
  - angular frequency $\omega = 2\pi\nu$, in radians per observation
- Example: monthly data with an annual cycle: $\nu = 1/12$ cycles per month, period 12 months
  - hourly data with a daily cycle: $\nu = 1/24$; quarterly data with a 6-year cycle: $\nu = 1/24$ as well
- Equivalent form: $A\cos(2\pi\nu t + \varphi) = a\cos(2\pi\nu t) + b\sin(2\pi\nu t)$
  - with $a = A\cos\varphi$, $b = -A\sin\varphi$ and $A^2 = a^2 + b^2$: linear in a and b, so it can be fitted by OLS
- The highest frequency visible in discrete data is $\nu = 1/2$ (one cycle every two observations): the **Nyquist frequency**

Two cycles plus noise



- Simulated series, $T = 120$: $x_t = 2\cos(2\pi t/12) + \cos(2\pi t/4 + 1) + \varepsilon_t$, $\varepsilon_t \sim N(0, 0.49)$ (Normal white noise with variance 0.49); right: its periodogram (defined in Section 3)

TSA_ch12_fourier

Interpreting two cycles plus noise

- In the time domain the two cycles are hard to separate by eye
  - in the frequency domain they are two isolated spikes, at $\nu = 1/12$ and $\nu = 1/4$
- Height of the spikes: 139.7 and 29.6; theory for a cycle of amplitude A at a Fourier frequency: $A^2 T/4$, i.e. 120 and 30
  - the spike grows with the squared amplitude: the larger cycle dominates
- The two spikes hold 85% of the total power; the noise spreads the rest evenly over all frequencies

Fourier frequencies and the discrete Fourier transform (1/2)

- For a sample $x_1, \ldots, x_T$, the **Fourier frequencies** are $\nu_j = j/T$, $j = 0, 1, \ldots, \lfloor T/2 \rfloor$
  - cycles that fit an integer number of times (j times) in the sample; $\lfloor T/2 \rfloor$: the integer part of $T/2$
- The **discrete Fourier transform** (DFT) measures how strongly the series moves with a cycle of frequency ν_j
  - $d(\nu_j) = T^{-1/2} \sum_{t=1}^{T} x_t \, e^{-2\pi i \nu_j t}$
  - i: the imaginary unit, $i^2 = -1$; $e^{-i\theta} = \cos\theta - i \sin\theta$ (Euler's formula)
  - so the DFT correlates the series with a cosine (real part) and a sine (imaginary part) of frequency ν_j

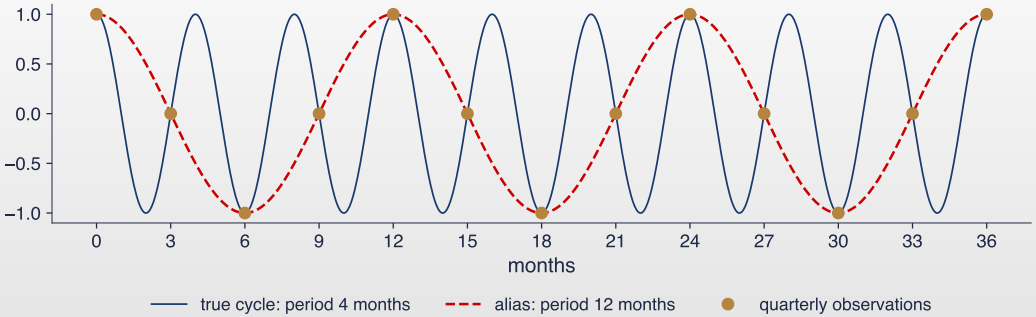
Fourier frequencies and the discrete Fourier transform (2/2)

- The T values $d(\nu_j)$ contain exactly the same information as the T data points
  - the data can be recovered from them by the inverse transform
- **Parseval**
  - the sum of squared deviations equals the sum of the squared moduli of the DFT
  - $\sum_{t=1}^{T}(x_t - \bar{x})^2 = \sum_{j=1}^{T-1} |d(\nu_j)|^2$
  - $\bar{x}$: the sample mean; $|d|^2 = (\mathrm{Re}\, d)^2 + (\mathrm{Im}\, d)^2$: the squared modulus of a complex number
  - reading: the variance of the series is split over the frequencies
- The **fast Fourier transform** (FFT) of Cooley and Tukey (1965) computes all $d(\nu_j)$ quickly
  - $O(T \log T)$ operations instead of $O(T^2)$: the number of operations grows like $T \log T$ instead of T^2

Worked example: the DFT of four numbers

- Data: $x = (4, 2, 0, 2)$, $T = 4$; mean $\bar{x} = 2$, deviations $y = (2, 0, -2, 0)$
  - Fourier frequencies: $\nu_1 = 1/4$ (period 4) and $\nu_2 = 1/2$ (period 2)
- At $\nu = 1/4$: $e^{-2\pi i t/4} = e^{-i\pi t/2}$ equals $-i$ for $t = 1$ and i for $t = 3$
  - $\sum_t y_t e^{-i\pi t/2} = 2(-i) + (-2)(i) = -4i$, so $d(1/4) = -4i/\sqrt{4} = -2i$ and $|d(1/4)|^2 = 4$
- At $\nu = 1/2$: $e^{-i\pi t} = (-1)^t$, so $\sum_t y_t(-1)^t = -2 + 2 = 0$ and $|d(1/2)|^2 = 0$
  - by symmetry $|d(3/4)|^2 = |d(1/4)|^2 = 4$
- Parseval: $4 + 0 + 4 = 8 = 2^2 + 0^2 + (-2)^2 + 0^2$; the whole variance ($8/4 = 2$) sits at period 4
- Check in Python: `abs(np.fft.fft([2, 0, -2, 0]))**2 / 4` gives $(0, 4, 0, 4)$

Aliasing



- A cycle of 4 months ($\nu = 1/4$ per month) observed only once a quarter (every 3 months)

TSA_ch12_fourier

Interpreting aliasing

- Per quarter, the true frequency is $3 \times 1/4 = 3/4$ cycles, above the Nyquist frequency $1/2$
  - it is folded back to $|3/4 - 1| = 1/4$ cycles per quarter: a period of 4 quarters, i.e. 12 months
- The quarterly points lie exactly on a 12-month wave: the data cannot tell the two cycles apart
  - this is **aliasing**: cycles faster than two observations appear as slower, false cycles
- Practical rule: sample at least twice per cycle of interest; average (rather than pick) values when you aggregate to a lower frequency

Recap: Cycles and frequencies

- A cycle has an amplitude, a frequency ν (cycles per observation), a period $1/\nu$ and a phase
- The DFT at the Fourier frequencies j/T rewrites the data as a sum of cycles; Parseval splits the variance by frequency
- Only frequencies up to $1/2$ are visible; faster cycles are aliased

From autocovariances to the spectrum

- For a stationary series with autocovariances $\gamma(h)$, $\sum_h |\gamma(h)| < \infty$, the **spectral density** is
  - $f(\nu) = \sum_{h=-\infty}^{\infty} \gamma(h) e^{-2\pi i \nu h} = \gamma(0) + 2\sum_{h=1}^{\infty} \gamma(h)\cos(2\pi\nu h)$, $-1/2 \le \nu \le 1/2$
  - $\gamma(h) = \mathrm{Cov}(x_t, x_{t+h})$: the autocovariance at lag h; $\sum_h |\gamma(h)| < \infty$: the autocovariances die out fast enough (short memory)
  - in words: $f(\nu)$ is a weighted sum of the autocovariances, with weights $\cos(2\pi\nu h)$ that oscillate at frequency ν
- The inverse relation: $\gamma(h) = \int_{-1/2}^{1/2} f(\nu) e^{2\pi i \nu h}\, d\nu$
  - the pair $\gamma \leftrightarrow f$ is the **Wiener–Khinchin** theorem: the autocovariances and the spectrum carry the same information
  - for $h = 0$: $\mathrm{Var}(x_t) = \gamma(0) = \int_{-1/2}^{1/2} f(\nu)\, d\nu$: the area under f is the variance
- Properties: $f(\nu) \ge 0$, $f(-\nu) = f(\nu)$; it is enough to draw f on $[0, 1/2]$
  - $f(\nu)\, d\nu$ is the share of the variance due to cycles with frequencies in $[\nu, \nu + d\nu]$

White noise, AR(1) and MA(1)

- **White noise** w_t, variance σ^2
  - $\gamma(h) = 0$ for $h \neq 0$, so $f(\nu) = \sigma^2$ for every ν
  - a flat spectrum: all frequencies contribute equally, like white light
- **AR(1)** $x_t = \phi x_{t-1} + w_t$, $|\phi| < 1$
  - $f(\nu) = \dfrac{\sigma^2}{1 - 2\phi\cos(2\pi\nu) + \phi^2}$
  - ϕ: the autoregressive coefficient; w_t: white noise with variance σ^2
  - $\phi > 0$: power at low frequencies (smooth, persistent series); $\phi < 0$: power at high frequencies (a zig-zag series)
- **MA(1)** $x_t = w_t + \theta w_{t-1}$
  - $f(\nu) = \sigma^2\,[1 + 2\theta\cos(2\pi\nu) + \theta^2]$
  - θ: the moving-average coefficient; $\theta > 0$ puts power at low frequencies, $\theta < 0$ at high frequencies
  - directly from the definition: $\gamma(0) = \sigma^2(1 + \theta^2)$, $\gamma(1) = \sigma^2\theta$, so $f = \gamma(0) + 2\gamma(1)\cos(2\pi\nu)$

The spectrum of an ARMA model

- **ARMA**(p, q)
  - $\phi(L)x_t = \theta(L)w_t$
  - L: the lag operator, $Lx_t = x_{t-1}$
  - $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$ and $\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$: the AR and MA polynomials
- Spectral density: the two polynomials evaluated at $z = e^{-2\pi i\nu}$
  - $f(\nu) = \sigma^2 \dfrac{|\theta(e^{-2\pi i\nu})|^2}{|\phi(e^{-2\pi i\nu})|^2}$
  - the MA part (numerator) adds power where $|\theta|$ is large; the AR part (denominator) adds power where $|\phi|$ is small
- A ratio of two polynomials in $\cos(2\pi\nu)$: smooth and easy to compute (Brockwell and Davis (1991), Sec. 4.4)

Worked example: AR(1) and MA(1) spectra

- AR(1) with $\phi = 0.6$, $\sigma^2 = 1$
  - $\nu = 0$: $\cos 0 = 1$, $f(0) = 1/(1-0.6)^2 = 1/0.16 = 6.25$
  - $\nu = 1/2$: $\cos \pi = -1$, $f(1/2) = 1/(1+0.6)^2 = 1/2.56 = 0.39$
  - ratio 16: slow cycles carry far more variance than fast ones
- MA(1) with $\theta = 0.5$, $\sigma^2 = 1$
  - $f(0) = (1+0.5)^2 = 2.25$ and $f(1/2) = (1-0.5)^2 = 0.25$
- Check of the area for the AR(1): $\int_{-1/2}^{1/2} f = \gamma(0) = 1/(1-\phi^2) = 1/0.64 = 1.5625$
  - the numerical integral of the formula gives the same value

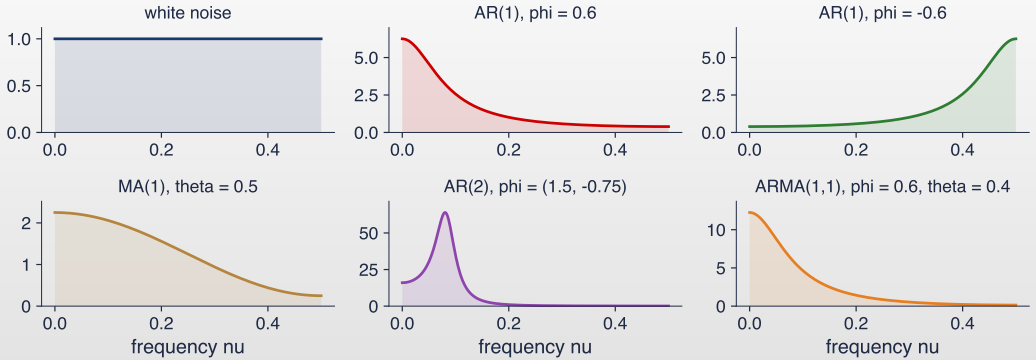
AR(2) and pseudo-cycles

- **AR(2)** $x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + w_t$
  - $f(\nu) = \dfrac{\sigma^2}{|1 - \phi_1 e^{-2\pi i \nu} - \phi_2 e^{-4\pi i \nu}|^2}$
  - with complex roots ($\phi_1^2 + 4\phi_2 < 0$) the ACF is a damped wave and the spectrum has a peak inside $(0, 1/2)$
  - the peak is at the frequency ν^* given by $\cos(2\pi\nu^*) = \phi_1(\phi_2 - 1)/(4\phi_2)$, when the right-hand side lies in $[-1, 1]$
- A **pseudo-cycle**
  - an irregular cycle whose length and amplitude vary from one round to the next
  - Yule (1927) fitted an AR(2) to Wolfer's sunspot numbers: the first autoregressive model, an alternative to fixed sine waves
- Business cycles are pseudo-cycles of this kind: recurrent, but never with the same length (Section 5)

Worked example: the peak of an AR(2) spectrum

- $\phi_1 = 1.5$, $\phi_2 = -0.75$, $\sigma^2 = 1$; complex roots, since $1.5^2 + 4(-0.75) = -0.75 < 0$
  - $\cos(2\pi\nu^*) = 1.5 \cdot (-1.75)/(-3) = 0.875$, so $\nu^* = \arccos(0.875)/(2\pi) = 0.0804$
  - period $1/\nu^* = 12.4$ observations; for yearly data, a cycle of about 12 years
- Height: $f(\nu^*) = 64$ against $f(0) = 1/(1 - 1.5 + 0.75)^2 = 16$
  - a clear peak: the series oscillates with a typical period of about 12 observations
- The same AR(2) is used in Section 4 to test the estimators of the spectrum

A gallery of spectral densities



- Theoretical spectra with $\sigma^2 = 1$ on $0 \leq \nu \leq 1/2$; the shaded area equals half the variance

TSA_ch12_spectral_densities

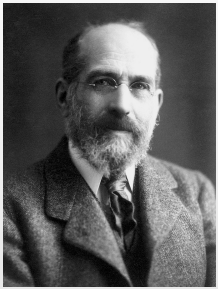
Interpreting the gallery

- Where the power sits tells the type of dependence
  - flat: no dependence (white noise); decreasing: positive persistence (AR(1) with $\phi > 0$, MA(1) with $\theta > 0$, ARMA(1,1))
  - increasing: alternation of signs (AR(1) with $\phi < 0$); a hump inside the band: a pseudo-cycle (AR(2))
- The AR(2) concentrates its variance 8.62 in a narrow band around $\nu = 0.0804$
  - the ARMA(1,1) adds the MA term to the AR(1): $f(0) = 12.25$, a steeper fall
- Reading direction: estimate the spectrum from data, look at its shape, then choose a model whose spectrum looks the same

Recap: The spectral density

- $f(\nu) = \sum_h \gamma(h) e^{-2\pi i \nu h}$; the area under f is the variance; f and the ACF are equivalent
- White noise is flat; AR(1) with $\phi > 0$ puts power at low frequencies; AR(2) with complex roots gives a peak
- ARMA spectra: $\sigma^2 |\theta(e^{-2\pi i \nu})|^2 / |\phi(e^{-2\pi i \nu})|^2$

Arthur Schuster and hidden periodicities



A. Schuster (1851–1934)

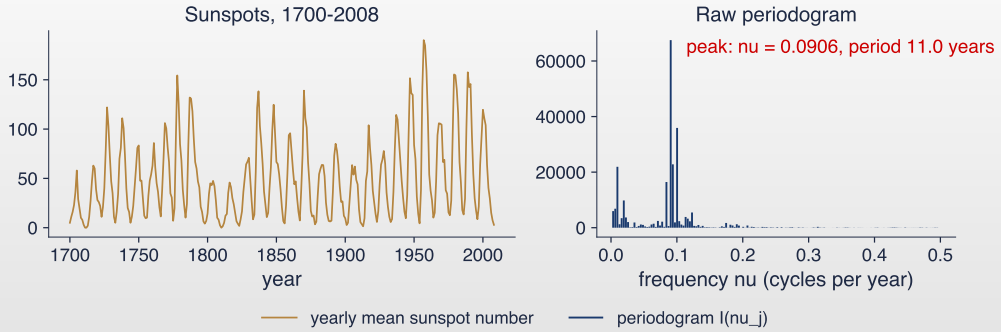
Photo: unknown author (1900s); public domain; Wikimedia Commons

- Schuster (1898) introduced the **periodogram** to search for "hidden periodicities" in meteorological data
  - a claimed 26-day cycle; he showed how to judge whether a peak could be due to chance
- Sunspots: counted daily since the 18th century; Rudolf Wolf built the standard index in 1848
  - their 11-year cycle is the classic test case of spectral analysis (Chapter 0)
- The question of the section: how do we estimate $f(\nu)$ from one sample, and how reliable is the estimate?

The periodogram

- **Periodogram**
  - $I(\nu_j) = |d(\nu_j)|^2$ at the Fourier frequencies $\nu_j = j/T$ (after removing the mean)
  - equivalently $I(\nu_j) = \sum_{|h|<T} \hat{\gamma}(h) e^{-2\pi i \nu_j h}$: the spectral formula with the sample autocovariances $\hat{\gamma}(h)$
- Regression view: regress x_t on $\cos(2\pi\nu_j t)$ and $\sin(2\pi\nu_j t)$ by OLS, with coefficients a_j, b_j
  - $I(\nu_j) = \frac{T}{4}(a_j^2 + b_j^2)$: the squared amplitude of the best-fitting cycle at ν_j, scaled by $T/4$
- Parseval (Section 1) in periodogram form: $\hat{\gamma}(0) = \frac{1}{T} \sum_{j=1}^{T-1} I(\nu_j)$
  - the share of a band of frequencies is the sum of its ordinates divided by the total
- In Python: `np.abs(np.fft.fft(x - x.mean()))**2 / T`; keep $j = 1, \ldots, \lfloor T/2 \rfloor$

Sunspots and their periodogram



⊡ Yearly mean sunspot number, 1700–2008, 309 observations (Wolf / SILSO series, statsmodels); right: the raw periodogram

TSA_ch12_periodogram

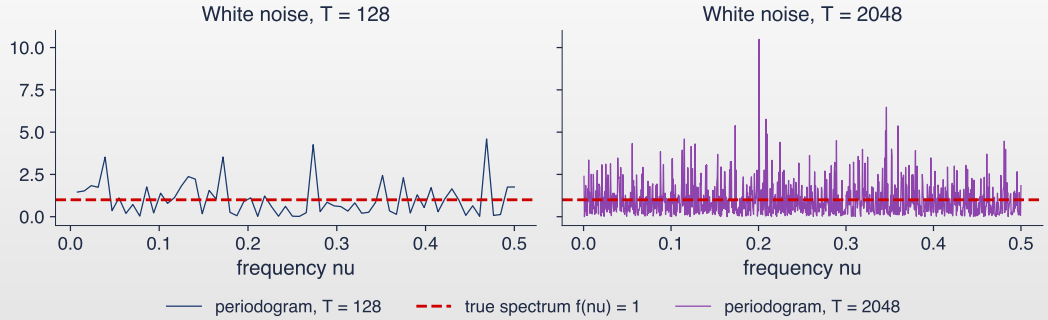
Interpreting the sunspot periodogram

- The highest ordinate is at $j = 28$: $\nu = 28/309 = 0.0906$ cycles per year, a period of 11.0 years
  - the next largest ordinates are at 10.0 and 10.7 years: the cycle is not exactly regular
- Periods between 9 and 13 years carry 60% of the variance
  - the small ordinates near $\nu = 0$ reflect slow changes in the height of the cycles (for example the weak cycles around 1800)
- The raw periodogram is very spiky: neighbouring ordinates jump up and down; the next slides explain why

Statistical properties of the periodogram

- For large T, at a Fourier frequency $0 < \nu_j < 1/2$ (Brockwell and Davis (1991), Sec. 10.3; Shumway and Stoffer (2017), Sec. 4.3):
  - $\frac{2I(\nu_j)}{f(\nu_j)} \approx \chi^2_2$: so $E[I(\nu_j)] \approx f(\nu_j)$ (unbiased) and $\mathrm{Var}[I(\nu_j)] \approx f(\nu_j)^2$
  - χ^2_2: the chi-square distribution with 2 degrees of freedom (mean 2, variance 4); $f(\nu_j)$: the true spectral density
  - ordinates at different Fourier frequencies are approximately independent
- The variance does **not** shrink when T grows
  - the periodogram is **not a consistent** estimator of f
  - a larger T gives more ordinates (a finer grid), not more precise ones
- $\chi^2_2/2$ is an exponential distribution with mean 1: a single ordinate can easily be 3 times the true value
  - its 95% quantile is 3.00: about 1 ordinate in 20 exceeds $3.00\, f(\nu)$ by chance

The periodogram of white noise



- Gaussian white noise with $\sigma^2 = 1$, so the true spectrum is flat at 1; samples of $T = 128$ and $T = 2048$

TSA_ch12_periodogram

Interpreting the white-noise periodogram

- $T = 128$: mean of the ordinates 1.03, standard deviation 1.03
  - $T = 2048$: mean 1.01, standard deviation 1.00: sixteen times more data, the same scatter
- The largest ordinate for $T = 2048$ is 10.5: with 1024 ordinates, isolated tall spikes appear by chance
  - a tall spike is not yet a cycle: it needs a test (next slide) or a smoothed estimate (Section 4)
- The mean is right (about 1): the periodogram is unbiased but noisy; averaging neighbouring ordinates will reduce the noise

Fisher's test of a hidden periodicity (1/2)

- Hypotheses
  - H_0: Gaussian white noise; H_1: white noise plus one cycle at an unknown Fourier frequency
- Statistic of Fisher (1929): the share of the largest ordinate in the sum of all ordinates
  - $g = \max_j I(\nu_j) \,/\, \sum_{j=1}^{m} I(\nu_j)$
  - m: the number of ordinates in $(0, 1/2)$; $\max_j$: the largest of them; $1/m \leq g \leq 1$
- Reading
  - under H_0 each ordinate is about $1/m$ of the total
  - a g much larger than $1/m$ signals a cycle: one frequency takes a large share of the variance

Fisher's test of a hidden periodicity (2/2)

- Exact p-value: the probability, under H_0, of a g above the observed value x
  - $P(g > x) = \sum_{k \geq 1,\ kx<1} (-1)^{k-1} \binom{m}{k} (1 - kx)^{m-1}$
  - k: the summation index, from 1 while $kx < 1$; $\binom{m}{k}$: the binomial coefficient
  - a small p-value (below 0.05): reject white noise in favour of a cycle
- Sunspots: $m = 154$, $g = 0.268$, against $1/m = 0.0065$; p-value < 0.001
  - the 11-year peak is far too large to be chance
- Caution: H_0 is white noise
  - against a red (AR(1)-like) background, low-frequency peaks are more likely by chance
  - so compare with a smooth background spectrum

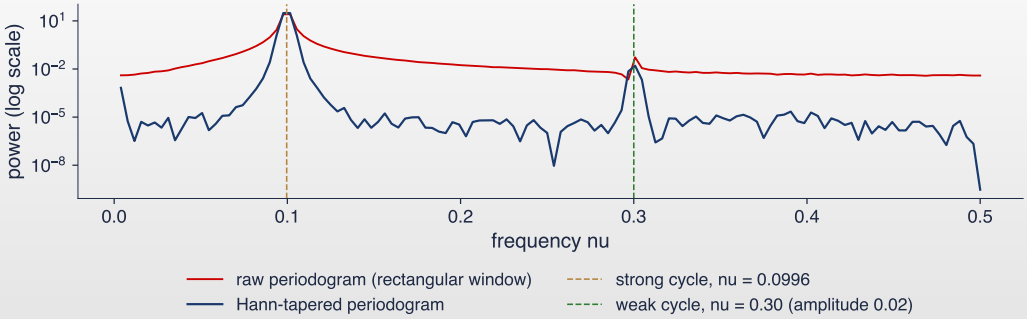
Recap: The periodogram

- $I(\nu_j) = |d(\nu_j)|^2$: proportional to the squared amplitude of the cycle at each Fourier frequency; it sums to the variance
- $2I/f \approx \chi^2_2$: unbiased but not consistent; neighbouring ordinates are nearly independent
- Sunspots: a peak at 11.0 years; Fisher's g rejects white noise

Leakage and tapering

- A cycle whose frequency is not a Fourier frequency does not fit a whole number of times in the sample
  - cutting the series at $t = 1$ and $t = T$ creates a jump; its power **leaks** into all other frequencies
  - leakage from a strong peak can hide a weak cycle elsewhere
- **Tapering**
  - multiply the data by a window h_t that goes smoothly to 0 at both ends before the DFT
  - Hann (cosine bell) window: $h_t = \frac{1}{2}[1 - \cos(2\pi(t - 0.5)/T)]$, rescaled to keep the total power
  - h_t is close to 0 at $t = 1$ and $t = T$ and equals 1 in the middle of the sample
  - the price: a slightly wider main peak (less resolution) for much smaller leakage
- Default in practice: always taper a little; always remove the mean (and a trend, if there is one) first

Leakage: raw and tapered periodograms



- $T = 256$: $\cos(2\pi \cdot 0.0996\, t)$ (between two Fourier frequencies) plus $0.02 \cos(2\pi \cdot 0.3\, t)$ and tiny noise; log scale

TSA_ch12_spectral_estimation

Interpreting leakage

- Raw periodogram: the strong cycle leaks power everywhere; at $\nu = 0.3$ the weak cycle stands only 7.7 times above the leaked background
  - with real noise added, this weak peak would be invisible
- Hann taper: the background falls by several orders of magnitude and the weak cycle stands 4 230 times above it
  - the main peak is a little wider: tapering trades resolution for less leakage
- Log scale is essential: on a linear scale both curves look like a single spike

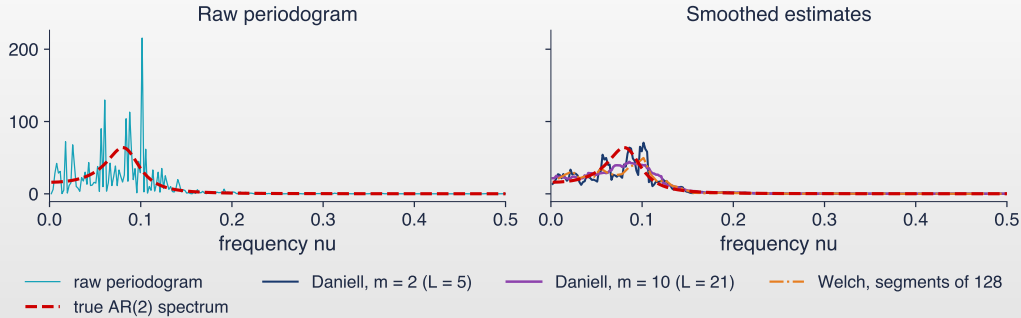
Smoothing the periodogram (1/2): the Daniell estimator

- Idea: f is smooth and the ordinates are nearly independent, so average $L = 2m + 1$ neighbouring ordinates
  - **Daniell** estimator: $\hat{f}(\nu_j) = \frac{1}{L} \sum_{k=-m}^{m} I(\nu_{j+k})$
  - m: the number of neighbours on each side; k: runs from $-m$ to m; $\hat{f}$: the estimated spectrum
  - variance about f^2/L: L times smaller than that of one ordinate
- **Bandwidth** $B = L/T$
  - the width of the frequency band that is averaged
- Bias–variance trade-off: a larger L lowers the variance but flattens narrow peaks (bias)
  - consistency needs $L \to \infty$ and $L/T \to 0$, e.g. $L \approx \sqrt{T}$

Smoothing the periodogram (2/2): lag windows and Welch

- Equivalent view: smoothing in frequency = down-weighting distant autocovariances (lag windows)
  - Bartlett window: $\hat{f}(\nu) = \sum_{|h|\leq M}(1 - |h|/M)\hat{\gamma}(h)e^{-2\pi i\nu h}$ (Bartlett (1950); Blackman and Tukey (1958))
  - M: the truncation lag; autocovariances beyond lag M are dropped, the others get weights $1 - |h|/M$ that fall linearly
  - a smaller M means more smoothing
- The **Welch** method (Welch (1967))
  - an average of the periodograms of K segments: $\hat{f}_W(\nu) = \frac{1}{K}\sum_{k=1}^{K} I_k(\nu)$, I_k: the tapered periodogram of segment k
  - split the series into K overlapping segments, taper each, average their periodograms
  - `scipy.signal.welch`; the segment length sets the resolution

Smoothed estimates of an AR(2) spectrum



- Simulated AR(2), $\phi = (1.5, -0.75)$, $T = 512$; true spectrum (dashed) with the raw periodogram (left) and three smoothed estimates (right)

TSA_ch12_spectral_estimation

Interpreting the smoothed estimates

- True peak at $\nu = 0.080$; the highest raw ordinate is at $\nu = 0.102$: a single noisy ordinate misplaces the peak
  - peak of Daniell $m = 2$: $\nu = 0.102$; of Daniell $m = 10$: $\nu = 0.086$; of Welch ($K = 7$ segments of 128): $\nu = 0.102$
- Mean squared error of $\log \hat{f}$: raw 1.86; Daniell $m = 2$: 0.25; $m = 10$: 0.09; Welch: 0.24
  - the price of heavy smoothing: the peak height drops from 64 to 44 (bias)
- Choose the bandwidth by looking at several values: keep the narrowest one whose estimate is no longer dominated by noise

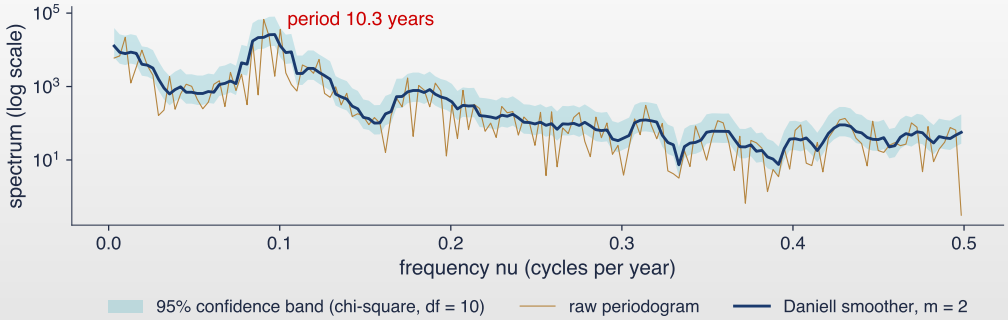
Confidence bands for the spectrum

- Daniell estimate with L terms: $\dfrac{2L\,\hat{f}(\nu)}{f(\nu)} \approx \chi^2_{2L}$ (degrees of freedom $df = 2L$)
  - 95% interval: $\left[\dfrac{df\,\hat{f}(\nu)}{\chi^2_{df}(0.975)},\ \dfrac{df\,\hat{f}(\nu)}{\chi^2_{df}(0.025)}\right]$
  - $\chi^2_{df}(p)$: the quantile of level p of the chi-square distribution with df degrees of freedom
- The interval is a fixed multiple of $\hat{f}$: on a log scale its width is the same at every frequency
  - a peak is significant if the band at the peak lies above the band of the background around it
- Welch with K non-overlapping segments: $df \approx 2K$; overlap and tapering change df a little

Worked example: a 95% band with $L = 5$

- $L = 5$ ($m = 2$), so $df = 10$; from the χ^2_{10} table: $\chi^2_{10}(0.025) = 3.25$ and $\chi^2_{10}(0.975) = 20.48$
  - lower bound: $10/20.48 = 0.49$ times $\hat{f}$; upper bound: $10/3.25 = 3.08$ times $\hat{f}$
- Sunspots: at the smoothed peak (period 10.3 years) $\hat{f} = 26\,095$
  - 95% interval: [12 740, 80 367]: asymmetric, wider upwards
- With only 5 ordinates the estimate is still uncertain by a factor of 2–3; more smoothing narrows the band but blurs the peak

Sunspots: smoothed spectrum with a confidence band



⊡ Daniell smoother with $m = 2$ ($L = 5$, $df = 10$, bandwidth $B = 5/309 = 0.016$ cycles per year) and its 95% band; log scale

TSA_ch12_spectral_estimation

Interpreting the smoothed sunspot spectrum

- The band at the peak (period 10.3 years) lies entirely above the band of the background at higher frequencies
  - the 11-year cycle is a robust feature, not a chance spike
- A second, smaller hump near $\nu \approx 0.18$ (about 5.5 years) is the first **harmonic**
  - the sunspot cycle is not a sine wave: it rises fast and decays slowly, and a non-sinusoidal cycle has power at multiples of its frequency
- Power also rises towards $\nu = 0$: the height of the cycles changes over decades and centuries

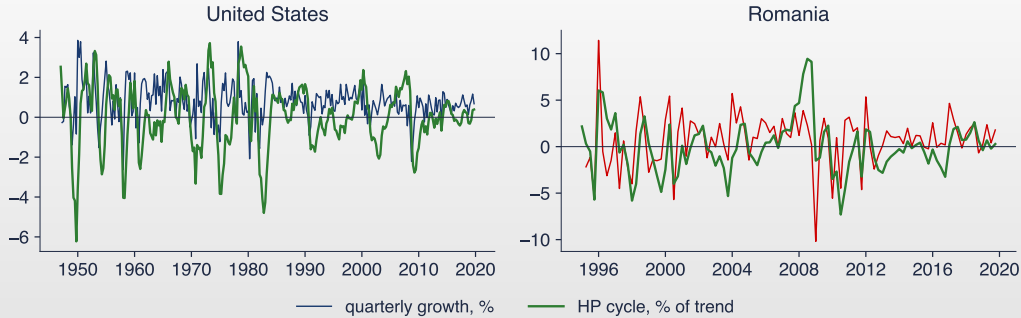
Recap: Estimating the spectrum

- Taper to reduce leakage; plot spectra on a log scale
- Smooth over L ordinates (Daniell, Bartlett, Welch): variance $\approx f^2/L$, but narrow peaks are flattened
- 95% band: $[df\ \hat{f}/\chi^2_{df}(0.975),\ df\ \hat{f}/\chi^2_{df}(0.025)]$ with $df = 2L$

The typical spectral shape and the business cycle

- Granger (1966): the spectrum of most economic levels (GDP, prices) falls steeply from $\nu = 0$: the **typical spectral shape**
  - trends and near unit roots put almost all the variance at the lowest frequencies and hide the cycles
  - so we first make the series stationary: growth rates (differences) or a detrended cycle
- **Business-cycle band**
  - fluctuations lasting 1.5 to 8 years, i.e. 6 to 32 quarters (Baxter and King (1999))
  - in frequency: $1/32 \leq \nu \leq 1/6$ cycles per quarter
- Data: US real GDP (FRED GDPC1, 291 quarterly growth rates) and Romanian real GDP (Eurostat, seasonally adjusted, 99 growth rates)
  - both up to 2019 Q4: the 2020 collapse and rebound would dominate every periodogram
  - the cycle is also extracted with the HP filter, $\lambda = 1600$ (Hodrick and Prescott (1997); Chapter 10)

GDP growth and HP cycles



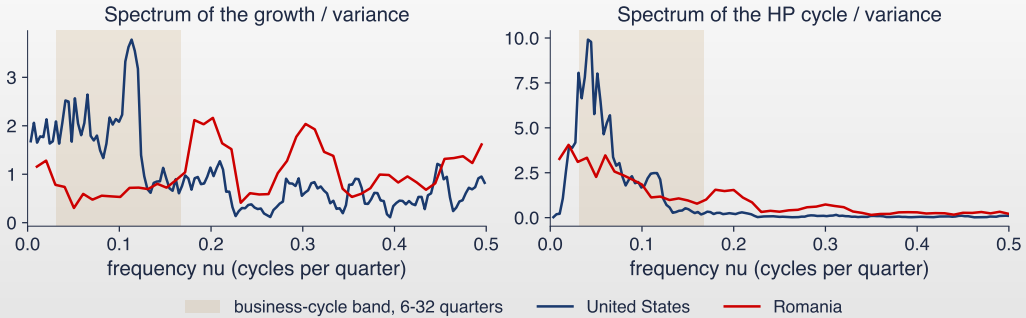
⊡ Quarterly growth $100\,\Delta \ln \mathrm{GDP}_t$ and the HP cycle ($100 \ln \mathrm{GDP}_t$ minus its HP trend), United States 1947–2019 and Romania 1995–2019

TSA_ch12_business_cycle

Interpreting the GDP series

- United States: mean growth 0.78% per quarter, standard deviation 0.93; HP cycle standard deviation 1.58% of trend
  - the cycle shows the recessions as deep troughs (1958, 1975, 1982, 2009)
- Romania: mean growth 0.73%, standard deviation 2.85: three times more volatile
  - one large boom–bust (2004–2010) dominates a sample of only 25 years
- The growth rate is noisy and short-lived; the HP cycle is smooth and persistent: their spectra will differ

Spectra of GDP growth and HP cycles



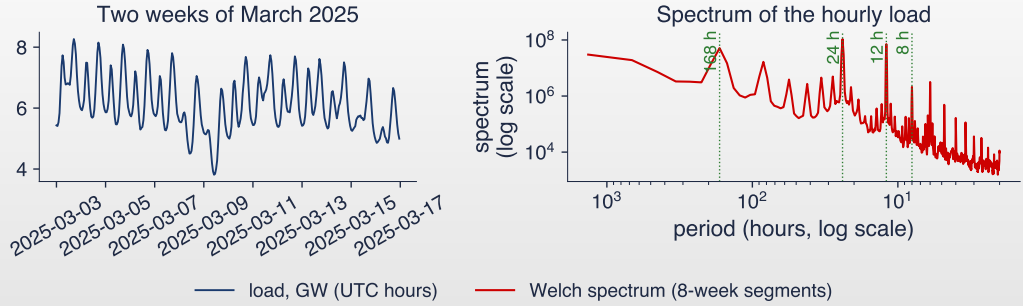
- Daniell smoother, $m = 2$; each spectrum divided by the variance of its series; shaded: periods of 6 to 32 quarters

TSA_ch12_business_cycle

Interpreting the GDP spectra

- US growth: 49% of the variance in the business-cycle band; the smoothed peak is at about 8.8 quarters
  - Romanian growth: only 16%; most of its variance is at high frequencies (quarter-to-quarter noise and revisions)
- US HP cycle: 80% of the variance in the band, peak at 24.3 quarters (about 6.1 years)
  - Romanian HP cycle: 49% in the band and 17% at periods longer than 8 years: one long cycle in a short sample
- Caution: the HP filter itself removes low frequencies and can shape the peak (Section 7); 99 Romanian observations give a very wide confidence band

The rhythm of electricity demand



- Hourly electricity load of Romania (ENTSO-E, MW, UTC hours), 2022–2026, 39 408 hours, mean 6 176 MW (Chapter 4); Welch spectrum with 8-week segments

TSA_ch12_electricity_load

Interpreting the load spectrum

- Sharp peaks at 24, 12 and 8 hours: the daily cycle and its harmonics (morning and evening peaks are not a sine wave)
  - the daily cycle and its harmonics hold 34% of the variance
- A peak at 167.7 hours (one week) and its harmonic at 84.0 hours: working days against weekends
  - weekly components: 16%; periods longer than a month (winter against summer): 33%
- This is why Chapter 4 modelled the load with daily and weekly Fourier terms: the spectrum shows which periods to include

Recap: Cycles in data

- Remove trends first (growth rates, detrended cycles): levels have the typical spectral shape
- US GDP: a business-cycle peak of about 6.1 years in the HP cycle; Romania: too short a sample for a sharp answer
- Electricity load: daily and weekly cycles with harmonics; the spectrum chooses the Fourier terms of a forecasting model

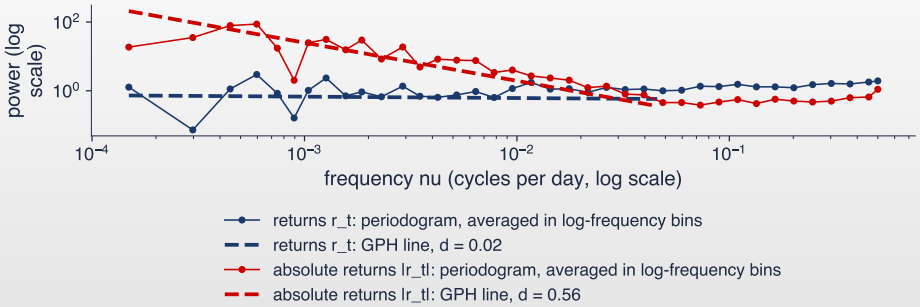
Long memory in the frequency domain (1/2)

- Chapter 8: a long-memory series has autocorrelations that decay like a power of the lag
  - $\rho(h) \sim Ch^{2d-1}$: $\rho(h)$ is the autocorrelation at lag h, $C > 0$ a constant, $\sim$ means "proportional for large h"
  - d: the memory parameter; $0 < d < 1/2$ gives stationary long memory; then $\sum_h \gamma(h) = \infty$ and $f(0) = \infty$
- ARFIMA$(0, d, 0)$: the spectrum explodes at zero
  - $f(\nu) = \sigma^2 \, |2\sin(\pi\nu)|^{-2d} \approx \sigma^2 (2\pi\nu)^{-2d}$ near $\nu = 0$: a **pole** at zero
  - taking logs: $\log f(\nu) \approx \log \sigma^2 - 2d \log(2\pi\nu)$, a straight line in $\log \nu$ with slope $-2d$
- Short memory (ARMA) has a finite $f(0)$: the log–log plot flattens at low frequencies

Long memory in the frequency domain (2/2): the GPH estimator

- Geweke and Porter-Hudak (1983): a regression of the log-periodogram on a log-frequency term
  - regress $\log I(\nu_j)$ on $-\log(4\sin^2(\pi\nu_j))$ for $j = 1, \ldots, m$; the slope is $\hat{d}$
  - standard error $\pi/\sqrt{24m}$: it shrinks as more frequencies are used
- Only the lowest $m = \lfloor T^{0.65} \rfloor$ frequencies are used, where the pole dominates
  - a larger m lowers the standard error but brings in short-memory effects (bias)
- Reading
  - $\hat{d}$ close to 0: short memory; $0 < \hat{d} < 0.5$: stationary long memory; $\hat{d} \geq 0.5$: nonstationary

The pole at zero in stock-market data



- Daily S&P 500 log returns, 2000–2026, $T = 6\,714$ (EODHD); periodogram averaged in 40 log-frequency bins; GPH lines on the lowest $m = 307$ frequencies (periods of 21.9 days and longer)

TSA_ch12_long_memory

Interpreting the log–log periodogram

- Returns r_t: flat at low frequencies, GPH $\hat{d} = 0.02$ (standard error 0.037): no long memory in the returns
  - a flat spectrum is what market efficiency predicts for returns (Chapter 1)
- Absolute returns $|r_t|$: power rises steadily towards $\nu = 0$, slope $-2\hat{d}$ with $\hat{d} = 0.56$
  - the long memory of volatility (Chapter 8); $\hat{d}$ near or above 0.5 also reflects breaks such as 2008 and 2020
- The same spectral picture explains why GPH and local Whittle (Chapter 8) are frequency-domain estimators

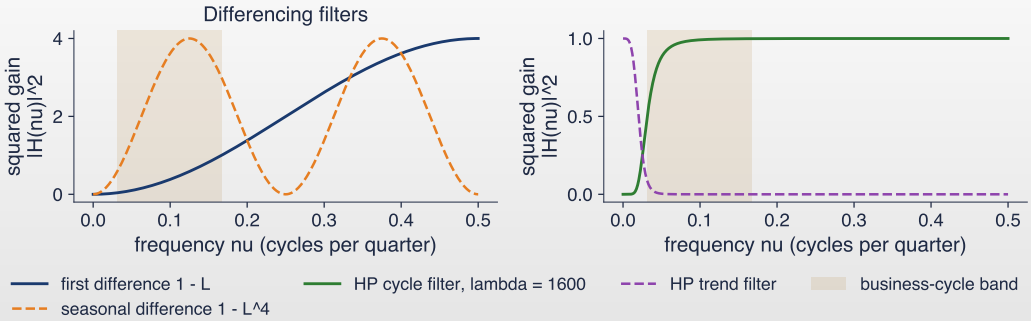
Linear filters and their gain (1/2)

- A **linear filter** turns an input series x_t into an output y_t by a weighted sum of current and past values
  - $y_t = \sum_j a_j x_{t-j}$; a_j: the filter weights; j: the lag
- Its **frequency response** is $A(\nu) = \sum_j a_j e^{-2\pi i \nu j}$
  - key result: $f_y(\nu) = |A(\nu)|^2 f_x(\nu)$; f_x, f_y: the spectra of input and output
  - $|A(\nu)|^2$ is the **squared gain**: above 1 the filter amplifies cycles of frequency ν, below 1 it damps them, 0 removes them

Linear filters and their gain (2/2)

- First difference $y_t = x_t - x_{t-1}$: $|A(\nu)|^2 = |1 - e^{-2\pi i\nu}|^2 = 4\sin^2(\pi\nu)$
  - zero at $\nu = 0$ (removes the trend), 4 at $\nu = 1/2$ (amplifies noise); at a period of 40 quarters: 0.025; at 4 quarters: 2
- Seasonal difference $(1 - L^4)x_t = x_t - x_{t-4}$ (quarterly data): squared gain $4\sin^2(4\pi\nu)$
  - zero at $\nu = 0, 1/4, 1/2$: it removes the trend and the seasonal cycles
- HP cycle filter: gain $G(\nu) = \dfrac{4\lambda(1 - \cos 2\pi\nu)^2}{1 + 4\lambda(1 - \cos 2\pi\nu)^2}$ (real), squared gain $G(\nu)^2$
  - λ: the smoothing parameter (1600 for quarterly data); a larger λ moves the cut-off to slower cycles
  - close to 0 at low frequencies, close to 1 at high ones: a **high-pass** filter

Squared gains of common filters



⊡ Left: first difference and seasonal difference ($s = 4$); right: HP cycle and trend filters, $\lambda = 1600$ (quarterly data); shaded: 6–32 quarters

TSA_ch12_filters

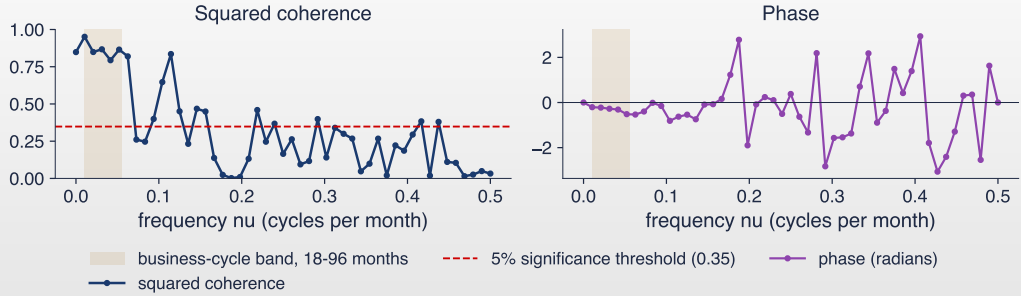
Interpreting the filter gains

- Differencing is not neutral: it weakens business cycles and amplifies the highest frequencies
  - this is why growth rates look noisier than levels; filtering white noise can even create spurious cycles (the Slutsky effect, Chapter 0)
- HP cycle filter: squared gain 0.996 at 8 quarters, 0.49 at 32 quarters, 0.24 at 40 quarters; half power at about 32 quarters
  - it keeps most of the power of cycles faster than about 8 years, including noise: not a band-pass filter (Baxter and King (1999))
- Applied to a random walk, the HP filter produces a cycle with a peak that comes from the filter, not from the data (Hamilton (2018))

Cross-spectrum, coherence and phase

- For two stationary series with cross-covariances $\gamma_{xy}(h) = \mathrm{Cov}(x_{t+h}, y_t)$:
  - **cross-spectrum** $f_{xy}(\nu) = \sum_h \gamma_{xy}(h) e^{-2\pi i \nu h}$, a complex number at each frequency
  - f_x, f_y: the spectra of x and y separately
- **Squared coherence**
  - $\rho^2_{xy}(\nu) = \dfrac{|f_{xy}(\nu)|^2}{f_x(\nu) f_y(\nu)} \in [0, 1]$
  - a squared correlation frequency by frequency: how well the cycles of x and y at ν move together
- **Phase** $\varphi(\nu) = \arg f_{xy}(\nu)$
  - the shift between the two cycles, in radians
  - arg: the angle of a complex number in the plane (its argument)
  - a time lag of k periods gives a phase that is linear in frequency: $\varphi(\nu) = \pm 2\pi\nu k$; so the lag is $|\varphi|/(2\pi\nu)$
- Estimation: smooth the cross-periodogram like the periodogram; the raw coherence is always 1, so smoothing is compulsory

Industrial production and unemployment



- US monthly data, 1948–2019 (FRED INDPRO, UNRATE), $T = 863$: $x_t = 100\,\Delta \ln \mathrm{IP}_t$ and $y_t = -\Delta u_t$ (the fall in the unemployment rate); Welch with $K = 8$ segments of 96 months

TSA_ch12_coherence

Interpreting coherence and phase

- In the business-cycle band (18–96 months) the squared coherence averages 0.87 (maximum 0.95), far above the 5% threshold 0.35
  - the 5% threshold for a Welch estimate with K segments: $1 - 0.05^{1/(K-1)}$
  - at frequencies faster than 6 months it averages only 0.17: the monthly noise of the two series is unrelated
  - the ordinary correlation, 0.49, mixes the strong link of the cycles with the weak link of the noise
- Phase at the 48-month cycle: -0.23 radians, a lag of about 1.7 months
  - the fall in unemployment follows the recovery of production with a short delay (Okun's law, with a lag)
- Where the coherence is low, the phase is meaningless: read the phase only in bands with significant coherence

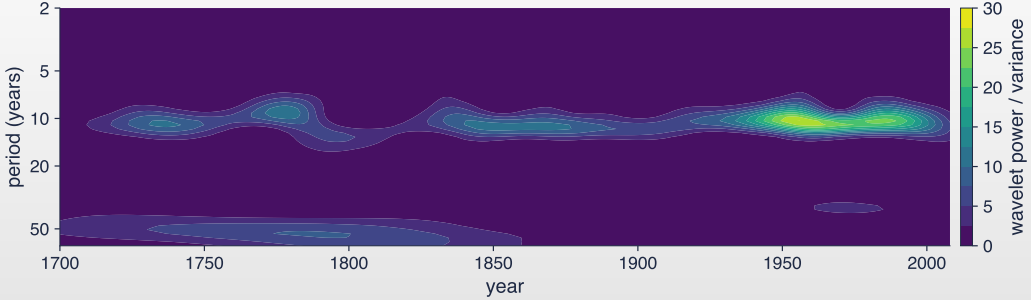
Wavelets (1/2): frequency that changes over time

- The spectrum assumes stationarity: the same cycles at every date
  - the **short-time Fourier transform** (STFT) computes spectra in moving windows of fixed length
- A **wavelet** is a short wave packet; the **continuous wavelet transform** (CWT) stretches it to each period and slides it along the series
  - the result, the **scalogram**, shows the power at each period and each date (Torrence and Compo (1998))
  - short windows for fast cycles, long windows for slow cycles: good resolution at all scales
- Beyond this course; a natural project topic (Python: `pywt`, or the Morlet code of this chapter)

Wavelets (2/2): the continuous wavelet transform

- The **Morlet wavelet**: a complex wave damped by a Gaussian bell
  - $\psi(u) = \pi^{-1/4}\, e^{i\omega_0 u}\, e^{-u^2/2}$, with $\omega_0 = 6$ (Torrence and Compo (1998))
  - u: time measured in units of the scale; ω_0: the number of oscillations inside the bell
- The **transform** at scale s and date τ
  - $W(s,\tau) = \sum_t x_t \, \frac{1}{\sqrt{s}}\, \psi^*\left(\frac{t-\tau}{s}\right)$: a correlation of the series with the wavelet stretched to scale s and centred at τ
  - ψ^*: the complex conjugate of ψ; for $\omega_0 = 6$ the period of the cycle is about $1.03\,s$
- The **scalogram**: the wavelet power $|W(s,\tau)|^2$
  - the analogue of the periodogram, but for each date τ separately

Wavelet power of the sunspots



Morlet wavelet power of the standardised yearly sunspot numbers, 1700–2008, periods 2 to 64 years; bright colours mean high power

TSA_ch12_wavelet

Interpreting the scalogram

- The 11-year band is present throughout, but its strength changes
  - average power in the 8–14-year band: 2.0 in 1795–1830 (the weak Dalton minimum) against 5.7 over the whole sample; the maximum is around 1956
- The periodogram averaged these changes into one peak; the scalogram shows when the cycle was strong or weak
- Edges of the chart are less reliable (the wavelet runs out of data): this region is called the cone of influence

Recap: Long memory, filters, two series, wavelets

- Long memory is a pole at $\nu = 0$: slope $-2d$ on a log–log plot (GPH)
- A filter multiplies the spectrum by its squared gain: differencing and the HP filter reshape cycles
- Coherence is a correlation by frequency; the phase gives the lead or the lag
- Wavelets show how cycles change over time

Possible contribution of AI

- **Code**
  - a first draft of a periodogram, a Daniell smoother, a Welch estimate or a coherence plot
- **Explanation**
  - a second explanation of leakage, aliasing or the bias–variance trade-off of the bandwidth
- **Exploration**
  - spectra of many series at once (load of several countries, GDP of all EU members) and a first reading of the peaks
- Example prompt
  - `Write Python code that reads hourly electricity load, removes the mean, computes the periodogram with a Hann taper and a Welch estimate with 8-week segments, plots both on a log-log scale against the period in hours, and marks 24, 12 and 168 hours.`

Checks you must run

- The units: frequency in cycles per observation or radians; `scipy.signal.welch` returns a one-sided density (twice our f)
- The period: $1/\nu$ in observations; convert to years, quarters or hours yourself
- Parseval: the ordinates must add up to T times the variance; test the code on a cosine with a known frequency
- A peak is a cycle only with a test or a confidence band; a raw periodogram always has tall spikes
- Trends, breaks and filters create low-frequency power and false peaks: check the data and the transformation first
- Every cited reference: it must exist; check the DOI

Key takeaways

- The spectrum splits the variance of a stationary series by frequency; it carries the same information as the ACF
- Read its shape: flat (white noise), falling (persistence), peaked (cycles), a pole at zero (long memory)
- The raw periodogram is unbiased but inconsistent: taper, smooth and add confidence bands
- Real cycles: 11 years in sunspots, about 6.1 years in the US HP cycle of GDP, 24 hours and 1 week in electricity load
- Filters reshape spectra; coherence and phase relate the cycles of two series

Key formulas

Quantity	Formula
Spectral density	$f(\nu) = \sum_h \gamma(h) e^{-2\pi i \nu h}, \quad \gamma(0) = \int_{-1/2}^{1/2} f(\nu)\, d\nu$
AR(1), MA(1)	$\sigma^2/(1 - 2\phi\cos 2\pi\nu + \phi^2), \quad \sigma^2(1 + 2\theta\cos 2\pi\nu + \theta^2)$
ARMA	$\sigma^2 \lvert\theta(e^{-2\pi i\nu})\rvert^2 / \lvert\phi(e^{-2\pi i\nu})\rvert^2$
Periodogram	$I(\nu_j) = \lvert T^{-1/2} \sum_t x_t e^{-2\pi i \nu_j t}\rvert^2, \quad \nu_j = j/T, \quad 2I/f \approx \chi^2_2$
Daniell	$\hat{f}(\nu_j) = \frac{1}{L} \sum_{\lvert k\rvert \le m} I(\nu_{j+k}), \quad L = 2m+1, \quad B = L/T$
95% band	$[2L\hat{f}/\chi^2_{2L}(0.975),\ 2L\hat{f}/\chi^2_{2L}(0.025)]$
Linear filter	$f_y(\nu) = \lvert A(\nu)\rvert^2 f_x(\nu), \quad \lvert 1 - e^{-2\pi i\nu}\rvert^2 = 4\sin^2(\pi\nu)$
Coherence, phase	$\rho^2_{xy} = \lvert f_{xy}\rvert^2/(f_x f_y), \quad \varphi = \arg f_{xy}, \quad \text{lag} = \lvert\varphi\rvert/(2\pi\nu)$

Self-assessment (1/2)

- **Question**: weekly data show a peak at $\nu = 0.0192$. What is the period?
  - **Answer**: $1/0.0192 \approx 52$ weeks: an annual cycle
- **Question**: what are $f(0)$ and $f(1/2)$ of an AR(1) with $\phi = -0.5$, $\sigma^2 = 1$?
  - **Answer**: $f(0) = 1/1.5^2 = 0.44$ and $f(1/2) = 1/0.5^2 = 4$: power at high frequencies
- **Question**: why does a longer sample not make the raw periodogram more precise?
  - **Answer**: each ordinate still has variance about f^2; more data only adds ordinates; precision needs averaging

Self-assessment (2/2)

- **Question**: a monthly series observed only every second month has a 3-month cycle. Where does it appear?
  - **Answer**: $\nu = 2/3$ cycles per two months is above $1/2$; it folds to $1/3$: a false cycle of 3 observations, i.e. 6 months
- **Question**: what does the first difference do to a cycle of 40 quarters?
  - **Answer**: it multiplies its power by $4\sin^2(\pi/40) = 0.025$: the cycle almost disappears
- **Question**: the log–log periodogram of a series has slope -0.6 near zero. What is d?
  - **Answer**: slope $= -2d$, so $d = 0.3$: stationary long memory (Chapter 8)
- **Question**: the squared coherence of two series is 0.9 at 5-year cycles and 0.1 at monthly cycles. What does it mean?
  - **Answer**: their business cycles move together; their short-run noise does not
- Next: Chapter 13, speculative bubbles and LPPL models

References (1/2)

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